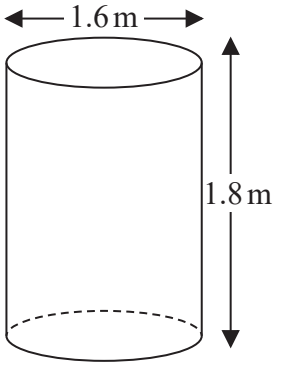
6 Jeremy has to cover 3 tanks completely with paint.

Each tank is in the shape of a cylinder with a top and a bottom.
The tank has a diameter of 1.6 m and a height of 1.8 m.

Jeremy has 7 tins of paint.
Each tin of paint covers 5 m^2

Has Jeremy got enough paint to cover completely the 3 tanks?
You must show how you get your answer.



Radius = 1.6 ÷ 2 = 0.8 m

Two circular faces: $2\pi r^2 = 2\pi(0.8)^2$

1 mark
Both circular ends.

Curved surface: $2\pi rh = 2\pi(0.8)(1.8)$

1 mark
Curved surface.

One cylinder = 13.069... m^2

1 mark
Area of one cylinder.

Three cylinders = 39.207... m^2

1 mark
Multiply by three.

7 tins cover 7 × 5 = 35 m^2; 8 cover 40 m^2.

No: he needs 8 tins.

1 mark
7 insufficient; 8 required.

(Total for Question 6 is 5 marks)